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Review Article

Research status and development prospect of Fe–Mn–C–Al system low-density steels

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ABSTRACT

The rapid development of automotive industry results in a series of increasingly serious problems such as energy consumption and environmental pollution, so that developing the automotive steels with high-strength and low-density is of great significance for energy conservation and emission reduction. Recently, Fe–Mn–C–Al system steel is favored by numerous researchers because of its low density and excellent combination of strength and ductility. However, too much Al addition inevitably triggers various changes of microstructure, deformation mechanism, and mechanical properties. In this paper, the composition design, production process, strengthening and strain hardening mechanism, and microstructure evolution are systematically expounded. The discussion mainly focuses on various strengthening (solid solution strengthening, grain refinement strengthening and precipitation strengthening) and strain hardening mechanisms (phase transformation-induced-plasticity (TRIP), twinning-induced-plasticity (TWIP), microband-induced-plasticity (MBIP), shear-band-induced-plasticity (SIP) and dynamic slip band refinement (DSBR), etc.). Also, the existing problems and future challenges in Fe–Mn–Al–C low-density steels are pointed out via analyzing the service performance (cryogenic performance, impact toughness, fatigue performance, hydrogen embrittlement, wear performance, and tensile properties). Finally, we look forward to the future development direction of Fe–Mn–Al–C low-density steels and provide the valuable ideas.

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With the rapid development of automobile industry, fuel efficiency and greenhouse gas emissions become increasingly serious. More and more advanced high-strength steel or lightweight materials (such as magnesium alloy, aluminum alloy, and carbon fiber materials) are employed in automobile production towards the above issues via achieving lightweighting. However, their expensive material cost and poor service properties (formability, wear, and hydrogen

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embrittlement et al.) result in a limited application in automotive materials.

Low-density steels, which are considered as the new generation automobile materials, have been developed to overcome the aforementioned shortages. As early as the 1930 s, Korter and Ton [1] first put forward the concept of Fe–Mn–C–Al system steels with Ni and Cr additions in consideration of the development prospect of high-strength steels. To lower the cost and density together, Ham and Carin [2] adopted the cheap Al and Mn elements instead of expensive Ni and Cr elements to optimize the chemical composition of stainless steels in the 1958 s. In the 1980 s and 1990 s [3–5], the replacement of Ni and Cr in stainless steel was widely studied, whereas few scientific studies on low-density steels were reported. Until the 2000 s, Frommeyer et al. [6] systematically investigated the microstructure and mechanical properties of Fe–Mn–Al–C low-density steels and considered that this steel can be employed to equip automobile steels. Since then, Fe–Mn–Al–C low-density steels started to attract much attention.

With the in-depth research of Fe–Mn–C–Al steels in recent years, it was found that Si addition presents a competitive advantage, which could restrain the carbide precipitation and enhance the mechanical properties [7–12]. However, some other scholars insisted that the Si addition played a negative role in improving the mechanical properties of this steel [13,14]. It can be seen that the role of Si element in Fe–Mn–Al–C steels is still controversial. In addition, Si element is not the main chemical component in Fe–Mn–Al–C low-density steels. Therefore, Si element will not be discussed in the following sections.

In recent years, the research on low-density high-strength steels mainly focuses on the composition design, role of alloying elements, and microstructure characteristics. The relationship between precipitates and matrix phases is also widely studied by numerous researchers. In Fe–Mn–Al–C steels, the variation of alloying element content results in the changes of both microstructures and strain hardening mechanisms, leading to the distant differences in application performance [13–18]. Therefore, the present work will give a brief introduction to alloying design and producing process. Furthermore, this work summarizes various strengthening and toughening mechanisms of single ferrite and austenite, austenite base duplex phase (A-BDP), and ferrite base duplex phase (F-BDP) steels with Fe–Mn–Al–C composition. Finally, the existing problems and future challenges under different service environments are investigated.

1. Composition design and microstructure characteristics of Fe–Mn–Al–C low-density steels

The composition design plays an important role in density and microstructure characteristics, such as lattice constant, phase composition and precipitate type. Thus, it is necessary to summarize the composition design and role of alloying elements, as well as the resulting influence on microstructure characteristics of Fe–Mn–C–Al low-density steels.

1.1. Composition design and role of alloying element

1.1.1. Density

In general, the density of Fe–Mn–Al–C steel is determined by alloying composition, as shown in Eq. (1) [19]:

$$\rho = 8.15 - 0.101\omega(\mathrm{Al}) - 0.41\omega(\mathrm{C}) - 0.0085\omega(\mathrm{Mn}) \quad (1)$$

where ω(Al), ω(C) and ω(Mn) represent the mass fractions of Al, C and Mn elements, respectively. Obviously, C content plays the most significant role in reducing density. However, the C solid solution is generally below 1.2 wt% in austenite of Fe–Mn–C–Al steel to restrain the carbide precipitation and crack generation during the plastic processing [20–22]. Therefore, the density of Fe–Mn–C–Al steel is mainly related to Al and Mn contents. Compared with the density of Mn (7.2 g/cm^3), that of Al (2.7 g/cm^3) has an obvious advantage in density reduction, implying that its low density shall be mainly controlled by Al content.

1.1.2. Lattice constant

During the thermal-mechanical processing, a large number of C, Mn and Al atoms are partitioned into austenite, resulting in the variation of austenite lattice constant. This complex relationship can be explained using Eq. (2) [23,24]:

$$\alpha = 0.35721 + 0.0002771\omega(\mathrm{Al}) + 0.0044089\omega(\mathrm{C}) + 0.0001166\omega(\mathrm{Mn}) \quad (2)$$

where ω(Al), ω(C) and ω(Mn) represent the mass fractions of Al, C and Mn elements, respectively. Meanwhile, C atom is much more important for expanding austenite lattice constant than those of Mn and Al atoms. Because C atom exists in the form of interstitial solid solutions, while Al and Mn atoms exist in the form of substitutional solid solutions.

It is reported that the larger mass fractions of C, Mn and Al elements within austenite of Fe–Mn–C–Al steel, the higher the yield strength, which can be explained via Eqs. (3)–(5) [25–28]:

$$\sigma_{yield} = 10.5\omega(\mathrm{Al}) + 351.5 \quad (3)$$

$$\sigma_{yield} = 0.58\omega(\mathrm{Mn}) + 328.0 \quad (4)$$

$$\sigma_{yield} = 180.7\omega(\mathrm{C}) + 250.7 \quad (5)$$

where σ_{yield} is the yield strength increment. It is obvious that C and Al play a more positive role in yield strength increment than that of Mn. The yield strength increment is positively related to the lattice constant variation which inevitably leads to the various solution strengthening effects.

1.1.3. Phase diagram

C, Mn and Al are the main elements for Fe–Mn–C–Al steels. Other micro-alloying elements (such as Si, Ni, Cr, V and Mo) also can be added to meet various service requirements. As the austenite stabler, C addition can improve the austenite stability and expand the austenite phase region [29]. Mn is considered another austenite stabler. When Mn content ranges from 5 to 12 wt%, austenite will first transform into ε-martensite and then ά-martensite. When Mn content is

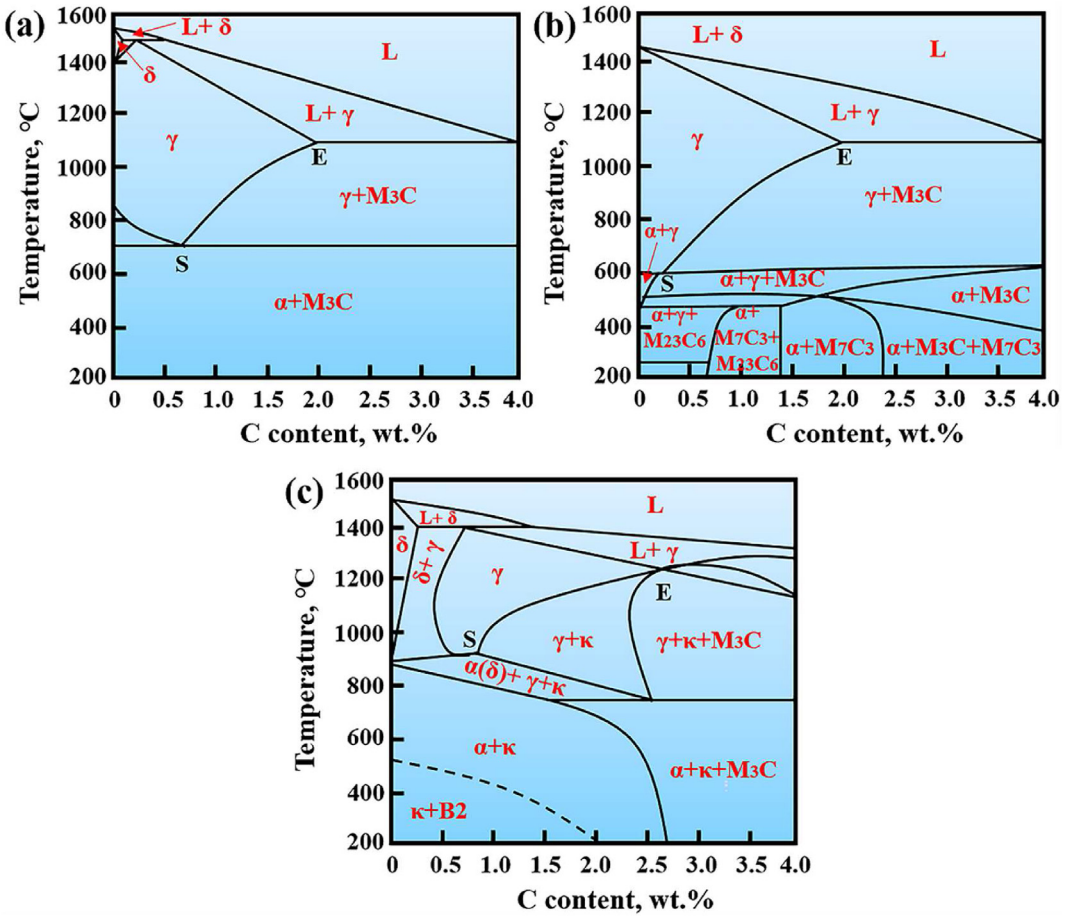


Fig. 1 – The effects of Mn and Al addition on phase constituent and fields of the Fe–C system [31–35]: (a) Fe–C, (b) Fe–15Mn–C, (c) Fe–7Al–C. The diashed line in (c) indicates the A2 to B2 transition temperature caused by Al addition.

located between 15 and 30 wt%, austenite will directly transform into martensite [19]. Al is the ferrite stabler. A high Al content can promote the ordered transformation of ferrite to (Fe, Mn)Al or (Fe, Mn)$_3$Al phases [30]. As the increasing Al content, the (Fe, Mn)$_3$C-type carbide in austenite will be replaced by (Fe, Mn)$_3$AlC-type carbide (κ-carbide).

Fig. 1 shows the influence of Mn and Al additions on phase constituents [31–35]. Fig. 1a shows the Fe–C phase diagram, the primary phases comprise of ferrite (α and δ), austenite (γ) and cementite (M_3C, M refers to Fe or Mn). Mn addition promotes the austenite phase region to expand to a lower temperature. The S-point temperature represents the lowest temperature at which austenite exists as a single phase, as shown in Fig. 1b. When the temperature is lower than S-point temperature, γ-austenite decomposes into α-ferrite and carbide, even forming M_7C_3 and $M_{23}C_6$ (Fig. 1b). In addition, increasing Mn content will result in the decreased C content and S-point temperature. A higher Mn content (>20 wt%) may accelerate the generation of β-Mn brittle phase (Fig. 1b). However, Al addition will make the (γ+M_3C) region replaced by (γ+κ) and (γ+κ+M_3C) regions, as shown in Fig. 1c. As the S-point temperature and C content increase, the (δ+γ) region expands and the γ single-phase region migrates to the right (Fig. 1c).

1.2. *Microstructure characteristics*

According to the alloy composition and main constituent phases, the Fe–Mn–Al–C system low-density steels after hot rolling and rapidly cooling to room temperature can be divided into single ferrite and austenite steels, A-BDP and F-BDP steels, as shown in Table 1.

Table 1 – Microstructural and compositional characteristics of low-density Fe–Mn–Al–C steels [36].

Fe–Mn–Al–C steels	single ferrite steel	single austenite steel	A-BDP steel	F-BDP steel
Alloy composition (wt.%)	$X_{Mn} < 5.0$ $5.0 < X_{Al} < 12.0$ $X_C < 0.1$	$15.0 < X_{Mn} < 30.0$ $2.0 < X_{Al} < 12.0$ $0.5 < X_C < 2.0$	$10.0 < X_{Mn} < 25.0$ $5.0 < X_{Al} < 12.0$ $0.6 < X_C < 1.0$	$3.0 < X_{Mn} < 10.0$ $3.0 < X_{Al} < 9.0$ $X_C < 0.4$
Constituent phases	δ, B2, DO3, κ	γ, α, δ, κ, β-Mn	α, B2, DO3, γ, κ, β-Mn	α, δ, B2, DO3, γ, κ, α′

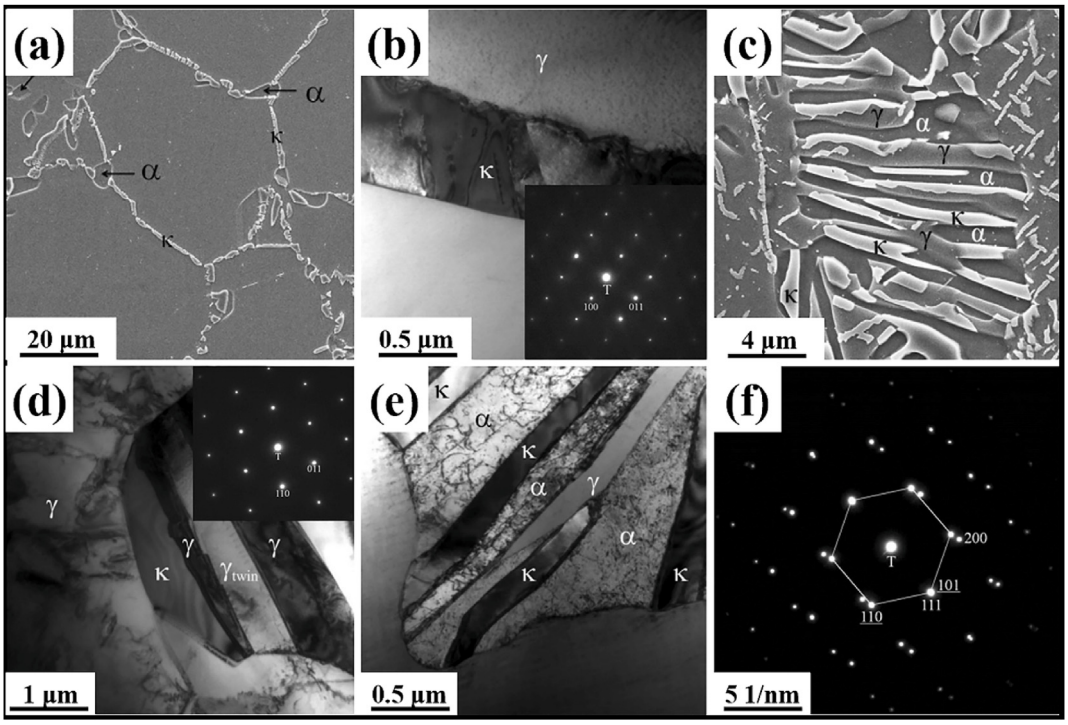


Fig. 2 – The distribution of κ-carbide at various locations [40]. (a, b) κ*-carbide precipitated along grain boundary; (c, d) κ′-carbide precipitated within grain; (e, f) κ-carbide precipitated simultaneously along grain boundary and within grain.

The room temperature microstructure of single ferrite Fe–Mn–Al–C steels is δ-ferrite. A high Al content results in an increase of ferrite recrystallization temperature. So the δ-ferrite grain is hard to be refined due to the limited dynamic or static recrystallizations [37]. Therefore, δ-ferrite prolongs along the rolling direction and presents the banded morphology under hot rolling conditions. During the subsequent cooling process, κ-carbide will precipitate along the δ-ferrite grain boundary if the cooling rate is slow. Moreover, Al content can significantly influence the crystal structures of matrix phases. Due to the difference in Al content, the disordered FeAl phase with A2, ordered FeAl phase with B2, or ordered Fe_3Al phase with DO3 structures can act as the matrix phases [38].

The room temperature microstructure of single austenite Fe–Mn–Al–C steels is composed of equiaxed austenite together with annealing twins [39], because the temperature of thermal-mechanical processing is generally located within the single austenite phase region. After cold rolling, the severe plastic deformation makes numerous grains elongated into lath-like morphology along the rolling direction. Due to the differences in element contents and cooling rates, the carbide types formed in austenite are κ-carbide and B2/DO3. Generally, κ-carbide can be divided into intragranular κ′-carbides and intergranular κ*-carbides, as shown in Fig. 2 [40].

The microstructure of Fe–Mn–Al–C duplex phase low-density steels is relatively complex, including austenite, ferrite (α and δ), B2, DO3, κ-carbide, β-Mn, and other precipitates (Table 1). The substructure, size, volume fraction and distributed location of these phases are different due to the influence of ferrite and austenite recrystallization [37,41] and element partitioning [18] under various heat-treatment processes. The quenched microstructure of A-BDP steels is full austenite. After aging at 500–800 °C, the lath-like ferrite will form along the austenite grain boundary, and the B2 phase is found within the lath-like ferrite. While the microstructures of hot-rolled and cold-rolled F-BDP steels are the typical bimodal banded structure of δ-ferrite and austenite. Meanwhile, κ-carbide is generated along the interface between austenite and δ-ferrite [42]. It can be seen that the evolution law of matrix phases caused by chemical composition has been relatively clear. However, a small difference in element content will lead to complex transformations of precipitate types, especially in A- and F-BDP steels. Thus, the depth-type purification during the smelting process is the primary technical problem to be solved.

2. Production process

Due to the different element contents in various Fe–Mn–Al–C low-density steels, the thermal-mechanical processing technologies, including plastic deformation and heat treatment, are also different. Generally, the forging ingot is homogenized at 1100–1250 °C for 1–3 h, then hot-rolled to 2–5 mm thickness between 1100 and 850 °C (Fig. 3). In some cases, it needs to reheat among different rolling-passes to avoid crack generation during hot rolling. A large rolling reduction can provide enough recrystallization driving force, thus inhibiting the segregation of Fe_3Al or FeAl. For austenite-base Fe–Mn–C–Al steels [19,43], the hot-rolled steels can be rapidly cooled to room temperature and then subjected to isothermal

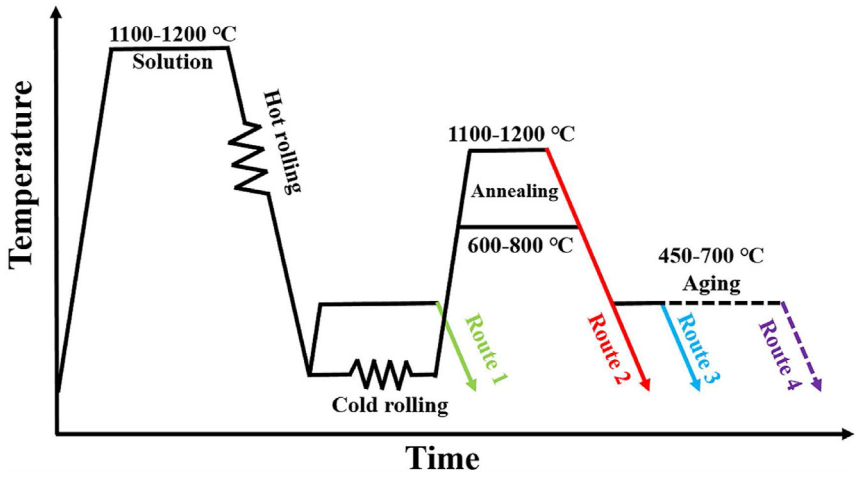


Fig. 3 – Abstract images of hot and cold rolling, annealing and aging processes of Fe–Mn–Al–C steel.

annealing and aging treatments (Route 1 in Fig. 3). However, the δ-ferrite in hot-rolled steels presents coarse and lath-like morphologies. Thus, cold rolling and recrystallization annealing are employed to tailor the δ-ferrite size, texture and κ-carbide (Route 2 in Fig. 3). Moreover, the cooling rate after hot rolling strongly influences intergranular κ-carbide formation [44]. Note that aging at 500–700 °C for 0.5–2 h (Routes 3 and 4 in Fig. 3) is beneficial to the nucleation of nano precipitates.

2.1. *Cold rolling and recrystallization annealing*

For Fe–Mn–C–Al low-density steels, the generation of deformation twins during cold rolling can effectively refine grains through the dynamic Hall-Petch effect to enhance strength [45]. However, the large cold rolling deformation also results in high residual stress, which harms the formability of steels [46]. The annealing process can eliminate work hardening and improve the toughness. Therefore, the combination of cold rolling and subsequent annealing process should be a reliable method to obtain high strength and good formability. Haase et al. [47] studied the influence of cold-rolling reduction (30%, 40% and 50%) and annealing time on the hardness of a Fe–22.5Mn–1.2Al–0.3C steel. The hardness was improved with the increase of cold rolling reduction due to the high work hardening but was decreased with the increase of annealing time due to recrystallization (Fig. 4). Zhang et al. [48] designed the A-BDP steel with ultrafine grains by adopting austenite recrystallization and ferrite transformation via annealing in a cold rolled Fe–25.7Mn–10.6Al–1.2C steel. The final microstructure consisted of recrystallized austenite, recovered austenite, ultrafine ferrite and nano κ-carbide, and the average grain size reached 1.12 ± 0.57 μm. An et al. [49] adopted a simple thermomechanical method of cold rolling and subsequent annealing to tailor the grain size and B2 particles in a Fe–28Mn–11Al–1C–5Ni steel. Nano B2 particles precipitated along the recrystallized austenite grain boundaries, while the ultrafine B2 particles were generated in coarse non-recrystallized austenite. Such a complex microstructure can significantly overcome the strength-ductility trade-off, but a few nano precipitates limit the enhancement of yield strength.

2.2. *Aging hardening*

Age hardening is a valid method to tailor the uniform distribution of carbide precipitates in austenite. Generally, the fine and dispersed carbides generate in a single austenite-base Fe–Mn–Al–C steel (Al>7 wt% and C > 0.7 wt%) after aging treatment, thus enhancing the precipitation strengthening. Chen et al. [50] studied the influence of aging temperature (500, 600 and 700 °C) on precipitate nucleation of the 1000 °C-annealed-water quenched Fe–30Mn–10Al–2C steel. After aging at 500 °C, a large number of κ-carbide precipitated along the austenite grain boundary. When aging was conducted at 600 °C, β-Mn was generated and showed dramatic secondary hardening. Further aging at 700 °C, the β-Mn was also observed along the austenite grain boundary, however, it played an inactive role in ductility. Liu et al. [51] investigated the influence of aging time (3, 10, 30 and 720 min) on carbide

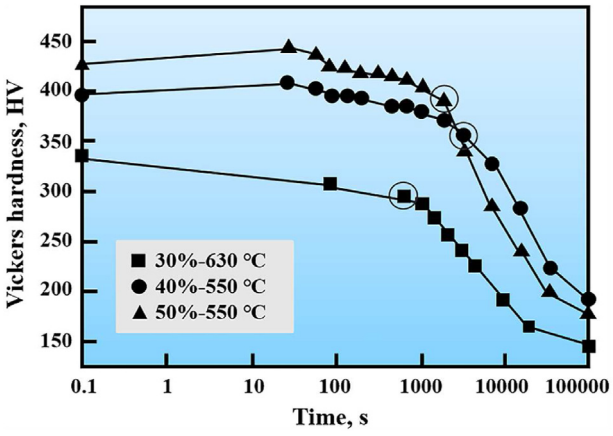


Fig. 4 – Vickers hardness evolution of 30%, 40% and 50% cold-rolled steels for different annealing times of Fe-22.5Mn-1.2Al-0.3C steel [47].

nucleation in the 1000 °C-annealed Fe–11Mn–10Al–1.2C steel. With aging at 500 °C for 3 min, prior κ-carbide quickly grew up and coarsened along the austenite grain boundary. After aging for 10 min, coarse κ-carbide nucleated along the initial austenite grain boundary, which decreased the C content of local region and further formed the C-depleted area, finally transforming into thin grain boundary ferrite. After aging for 30 min, austenite within the C-depleted area decomposed into ferrite and κ-carbide via eutectoid reaction. When the aging time reached 720 min, the continuous eutectoid reaction increased the fraction of nano κ-carbide precipitates, finally contributing to a high yield strength. Bidyapati et al. [52] pointed out that the proper aging time could accelerate the intragranular B2 phase and κ-carbide precipitation at once. However, a long aging time resulted in the decreased fractions of B2 phase and κ-carbide. In addition, aging process also can be carried out after cold rolling to enhance mechanical properties. Hwang et al. [53] studied the influence of aging treatment on a Fe–21Mn–10Al–1C–5Ni steel under partial and complete recrystallization annealing conditions. The result showed that the hardness of B2 phase and yield strength were significantly improved after the annealing-aging process. So, the combination among cold rolling, recrystallization annealing, and aging processes can not only ensure a high yield strength but also overcome the strength-ductility trade-off in Fe–Mn–Al–C steels.

2.3. *Other processes*

To pursuit more outstanding combination between strength and ductility, the scholars adopted some other thermal mechanical processing technologies. Ren et al. [54] adopted the rapidly water-quenched to manufacture the short-range ordering phase and κ-carbide in a Fe-30.4Mn–8Al-1.2C steel. As the cooling rate increased, the transformation from short-range ordering to κ-carbide was restrained, which increased the work hardening without deteriorating yield strength. Kwang and Hyo et al. [55] developed a low-temperature tempering-induced partitioning process to overcome the strength-ductility trade-off in a Fe-2.8Mn-5.7Al-0.3C steel. The austenite grains with heterogeneous size-distribution were embedded in ferrite, which provided a suitable condition for C partitioning. Interestingly, more C preferentially partitioned into smaller austenite grains than larger ones. This size-dependent partitioning led to the slip spacing variation and lattice strain under dislocation motion.

In the above, many factors influence the microstructures and mechanical properties of Fe–Mn–Al–C low-density steels. Table 2 summarizes the mechanical properties of various forged plates obtained under different thermal-mechanical processes. It can be seen that the tensile strength and total elongation of low-density steels strongly depend on cold rolling-annealing and aging hardening. Because the type, distribution and size of precipitates are susceptible to annealing/aging temperatures and times. Thus, it is necessary to precisely control the type, size and distribution of precipitates, which requires relevant scholars to optimize the existing process and to further narrow the temperature and time intervals on the basis of clarifying precipitation rules.

3. Strengthening and strain hardening mechanisms

The strengthening and toughening mechanisms determine the mechanical properties of steels. For single ferrite and austenite Fe–Mn–Al–C low-density steels, solution strengthening, fine grain strengthening, dislocation strengthening and strain hardening are the main strengthening and toughening mechanisms [19]. While that of dual-phase Fe–Mn–Al–C low-density steels mainly includes TRIP and TWIP effects in addition to the above mechanisms, resulting in the multi-stage strain hardening characteristics due to the synergetic effect of various deformation mechanisms [20].

3.1. *Strengthening mechanisms*

3.1.1. *Solution strengthening*

Austenite-base Fe–Mn–Al–C steels contain a large fraction of Al, Mn and C elements. Interstitial C and N atoms enter into FCC γ-Fe, the lattice distortion is caused and the stress field is formed simultaneously, which then interacts with dislocations in the matrix to generate a strong hardening effect. In contrast, the lattice distortion caused by the replacement atoms (such as Mn, Ni or Si) is spherical symmetry, so its strengthening effect is weaker than that of interstitial solid solution atoms [26,56]. The influence of alloy elements on yield strength of austenite steels is shown in Fig. 5 (a). The 1 wt % C contributes to a yield strength increment of 187–300 MPa for austenite, while the contributed yield strength of 1 wt% Al

Table 2 – Comparison of mechanical properties of low-density steels under different thermal-mechanical processes. TE and UTS represent total elongation and ultimate tensile strength, respectively.

Composition (wt.%)	Density ($g \cdot cm^{-3}$)	TE (%)	UTS (MPa)	Thermal-mechanical processes
Fe–15Mn–10Al-0.86C–5Ni [19]	6.71	18.0	1548	Cold rolling-annealing
Fe–23Mn-1.5Al-0.3C [48]	6.98	98.9	840	Hot rolling-solution
Fe–27Mn–12Al-0.9C [24]	6.46	40	800	Cold rolling-water quenching
Fe–11Mn–10Al-1.2C [51]	6.5	22.2	1314	Cold rolling-annealing-aging
Fe–30Mn–11Al-1.2C [34]	6.29	17.8	1218	Solution-aging
Fe–21Mn–10Al–1C–5Ni [54]	7.0	80	775	Solution-water quenching
Fe–29Mn-9.4Al-0.3C [47]	7.4	30	757	Cold rolling-annealing
Fe–28Mn–9Al-0.9C–5Ni [53]	6.79	20.3	1620	Cold rolling-annealing-aging

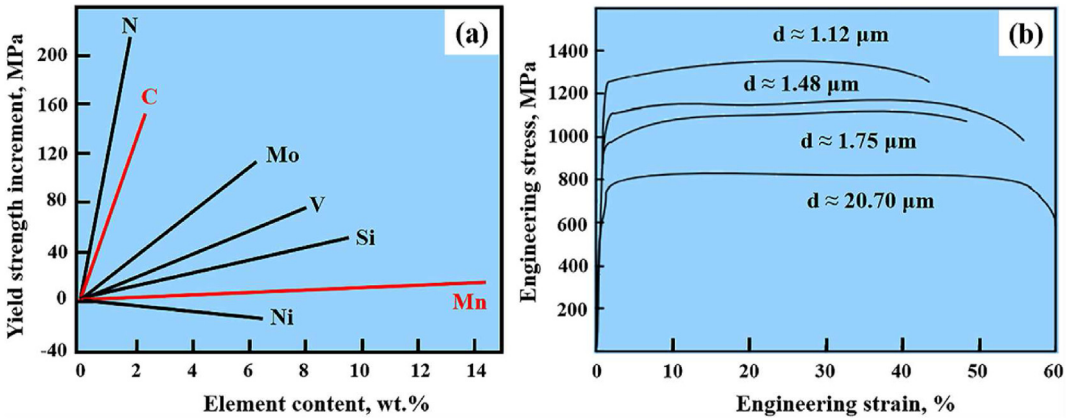


Fig. 5 – (a) Contributed yield strength increment of alloying elements in austenite steel [6]; (b) Engineering stress-strain curves of samples with various austenite grain sizes [48].

is slightly below 10 MPa. Mn addition can not only improve the austenite fraction but increase the solubility of Al and C in austenite, both of which indirectly affect the mechanical properties.

3.1.2. *Grain refinement strengthening*

According to the Hall-Petch effect, grain refinement strengthening is another strengthening mechanism. Generally, cold rolling, recrystallization annealing, and adding micro-alloying elements (Nb, V, Ti, and Mo et al.) are also used to refine grains [55,57–59]. Zhang et al. [48] prepared a cold-rolled Fe–Mn–Al–C dual-phase steel with ultrafine grains in consideration of austenite recrystallization and ferrite transformation. The result implied that the austenite grain refinement increased the yield strength and ultimate tensile strength above 400 MPa (Fig. 5b). However, too small grains will weaken this strengthening effect.

3.1.3. *Precipitation strengthening*

Fe–Mn–Al–C low-density steels included numerous precipitates, such as κ-carbide, B2, DO3 and β-Mn. Due to the different precipitate modes, the corresponding precipitation strengthening can be divided into two types. The first one is related to the uniformly distributed nano κ-carbide, which affects the movement and arrangement of dislocations during plastic deformation [60]. While the other one is related to elongated α-ferrite, which can continuously and orderly transform into B2 or DO3 intermetallic compounds during the aging process [61]. However, β-Mn can significantly deteriorate the ductility of Fe–Mn–Al–C low-density steels and reduce impact energy, it will not be described in detail [20,62].

(1) κ-Carbide strengthening

The κ-carbide has L′12 ordered crystal structure similar to that of perovskite oxide, as shown in Fig. 6 [63]. The chemical formula of κ-carbide is usually $(Fe, Mn)_3AlC_x$. Still, the exact chemical composition is unknown because the C content in ultrafine precipitates is challenging to measure. In 1995, Palm [64] reported that the chemical formula of κ-carbide should be $(Fe, Mn)_yAl_{1-y}C_x$ ($-0.2 < y < +0.2$, $0.42 < x < 0.71$) through experiment and accurate calculation. Although the commonly used Fe_3AlC formula does not take into account the Fe/Al ratio deviation, it is still widely used. Seol et al. [42] proposed that the formation of κ-carbide was determined by a eutectoid reaction related to nucleation and growth, which further revealed that the composition of κ-carbide varied with the aging conditions and alloy composition.

High C (>0.7 wt%) and Al (>7.0 wt) contents are the primary conditions for κ-carbide formation in single austenite-base Fe–Mn–Al–C steels. However, κ-carbide can be divided into κ′-carbide and κ^*-carbide according to the aging temperature. When the aging temperature ranges from 500 to 700 °C, the spinodal decomposition will occur in austenite, which is described as follows: $\gamma \rightarrow \gamma' + \gamma'' \rightarrow \gamma' + L1_2$ (short-range ordering phase) $\rightarrow \gamma' + \kappa'$ [65]. Meanwhile, the κ′-carbide refers to the ultrafine precipitate nucleated within the austenite matrix (Fig. 6), showing a coherent relationship between them [29,42,61]. Finally, the ultrafine layer κ′-carbide in austenite and nano dispersed κ′-carbide in ferrite can significantly enhance yield stress and cause aging hardening [66,67]. As the aging temperature increases to 700–900 °C, the coarse κ′-carbide will generate in austenite matrix or along the prior austenite grain boundary. This κ′-carbide is called intercrystalline κ^*-carbide. Therefore, the high aging temperature can accelerate the precipitation of coarse κ^*-carbide along the grain boundary. Moreover, the long aging time will lead to the formation of brittle β-Mn phase. However, κ^*-carbide and β-Mn are unfavorable to the ductility and toughness of Fe–Mn–Al–C low-density steels [68]. Therefore, the aging temperature and time should be precisely regulated to avoid the formations of intergranular κ^*-carbide and β-Mn in single austenite-base Fe–Mn–Al–C steel. Under a plastic deformation, the interaction between κ-carbides and dislocations does not change the plane slip mode, but the additional dislocation sources are activated and acted upon by cross slip [38]. Moreover, these nano κ-carbides uniformly distribute in austenite and then promote the plane slip. As the strain increases, the plane dislocation structure is further refined, contributing to continuous strain hardening. This is the main contribution of κ-carbide strengthening in single austenite-base Fe–Mn–Al–C steels.

(2) B2 phase strengthening

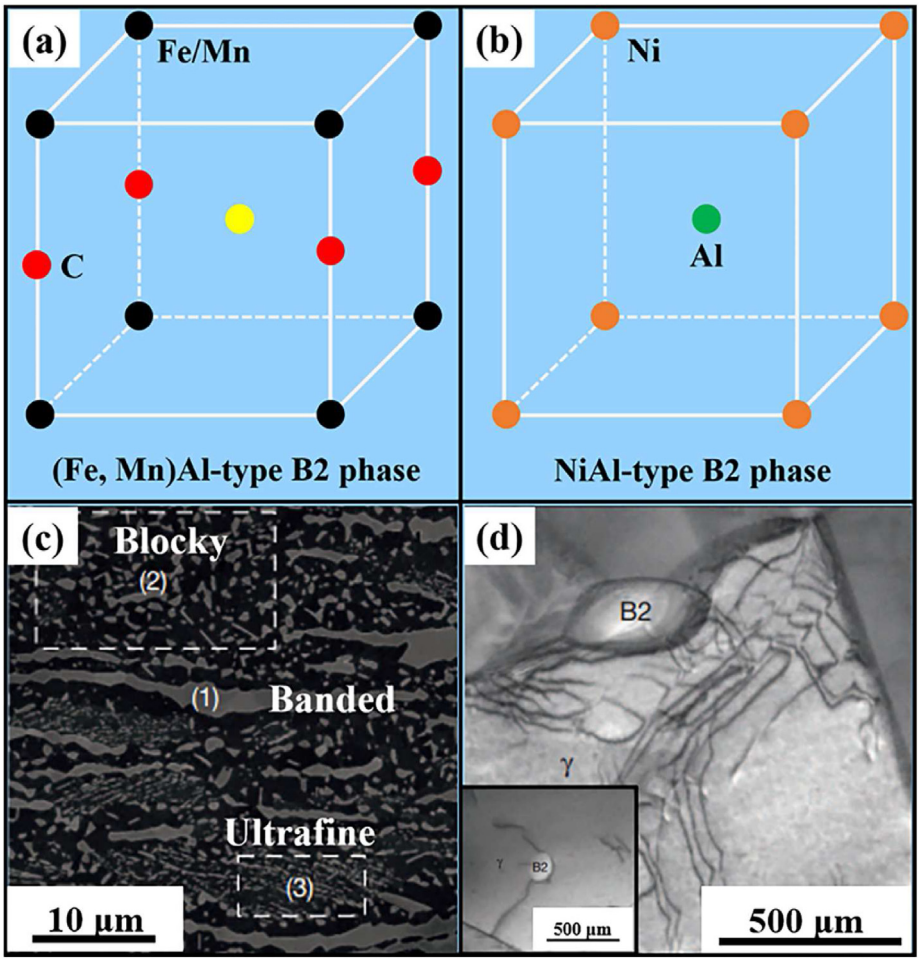


Fig. 6 – (a, b) Lattice of B2 phase and NiAl ordered phases in Fe–Mn–Al–C low density steel [69]; (c, d) B2 phase precipitates in aged Fe–Mn–Al–C steel [71].

When the Fe–Mn–Al–C low-density steels undergo quenching-aging/cold rolling-annealing, the (Fe, Mn)Al or NiAl type ordered intermetallic compounds are formed. B2 phase is an ordered body-centered cubic structure of CsCl type, as shown in Fig. 6a and b [69]. The previous reports imply that B2 at room temperature is unfavorable to plasticity, but it is favorable to corrosion resistance, oxidation resistance and wear resistance [70].

In addition to κ-carbide, a small amount of ultrafine α-ferrite precipitate along the austenite grain boundary in A-BDP Fe–Mn–Al–C steels under different annealing temperatures. High Al content can induce the α-ferrite transformation into B2 and DO3. Kim et al. [71] reported that the Ni addition in Fe–Mn–Al–C steels effectively combined with Al to form the NiAl-type B2 phase. Numerous studies implied that cold rolling and annealing processes could provide a suitable condition for B2 nucleation [42,43,53]. After cold rolling and annealing, it was found that there are three different morphologies (banded, blocky and ultrafine) of B2 phases (Fig. 6c) in the Fe–16Mn-9.6Al-0.86C-4.9Ni steel [71]. The banded B2 phase was transformed from δ-ferrite (white rectangle area). The blocky B2 phase, which precipitated along the recrystallized austenite grain boundaries, was transformed from α-ferrite (yellow rectangle area). However, the ultrafine B2 particles nucleated on shear bands of recrystallized austenite grains (red rectangle area). The widely distributed B2 phase contributed to the enhancements of yield strength, tensile strength, and ferrite hardness.

Compared with κ-carbide, B2 phase is not sheared by slip dislocations. This phenomenon leads to the dislocation pile-up at phase boundary, finally enhancing the work hardening of Fe–Mn–Al–C steels containing B2 phase during plastic deformation (Fig. 6d) [71]. In this regard, relevant scholars have given different opinions. Yang et al. [72] believed the enhancement of work hardening was attributed to high back stress due to the presence of B2 phase. During plastic deformation, the strain incompatibility between the austenite matrix and B2 phase accelerated the dislocation multiplication at phase interfaces and formed high internal stress, resulting in the retardation of dislocation slip in austenite. At a much larger plastic deformation, B2 phase began to yield and the long-term back stress was formed. However, Kim et al. [73] insisted that the short-range ordering phases were the internal factor responsible for work hardening, rather than the difference in stress/strain distribution in Fe–Mn–Al–C steels. Sohn et al. [74] pointed out that the NiAl-type B2 phase precipitated in Ni-containing steels had a stronger chemical bond than FeAl-type B2 precipitates in Ni-free steels, which

was conducive to improving strength. However, a large number of hard and brittle B2 phases distributed along the grain boundary would deteriorate the plasticity and toughness.

In addition, Cu is added into single austenite-base Fe–Mn–Al–C steels to induce the B2 phase formation [75]. Cu, as an austenite stabler, can increase the volume fraction of austenite and delay the recrystallization, due to the drag effect [76,77]. Song et al. [78] added 0.3 wt% Cu into a Fe–12Mn–7Al-0.5C dual-phase steel to form a large number of nano Cu-rich B2 particles. The plane slip and ultrafine dislocation substructure (Taylor lattice) were observed during subsequent plastic deformation. The synergistic effect among solution hardening, delayed recrystallization, and Cu-rich B2 particles significantly enhanced the yield strength and strain hardening.

(3) MC carbide strengthening

In recent years, the addition of micro-alloying element (Nb, V and Ti et al.) has been widely employed to improve the mechanical properties of Fe–Mn–Al–C low-density steels. Due to a high affinity between micro-alloying (Nb, V and Ti et al.) and C atoms, numerous stable MC carbides are formed, effectively reducing the nucleation driving force of other carbides, thus inhibiting the precipitation and growth of other carbides [57,58]. Song et al. [60] considered that the yield strength increment from MC carbides was similar to that from κ-carbide in Fe–Mn–Al–C steels under suitable microstructure controlling. Zargaran et al. [59] confirmed that the 0.2 wt% Nb addition into a Fe–8Al–5Mn-0.1C steel could result in 0.2% volume fraction of NbC precipitates, much lower than 2.3% volume fraction of κ-carbide. Although NbC precipitates have poorer inhibition on κ-carbide precipitation, the austenite grain size is obviously refined. Compared with Nb-free steels under the same conditions, the yield strength of Nb-containing steel increases by about 100 MPa without deteriorating ductility. However, a few scholars hold different opinions on the above conclusion. Li et al. [79] studied the effect of Ti, Mo and Nb composite additions on microstructure and tensile behavior of a Fe–26Mn–8Al–1.5C steel. It was found that the additions of Ti, Mo and Nb could refine the grains, increase the size and volume fraction of κ-carbide, and form intercrystalline and intragranular (Ti, Mo, Nb)C particles. The coarse κ-carbide and nano (Ti, Mo, Nb)C particles on grain boundary resulted in the poor ductility. It can be seen that there is no agreed theory about the effect of MC carbide on κ-carbide precipitation and growth, and further research is needed.

Hereby, the Mn and C contents can affect the solution strengthening, precipitation strengthening and austenite characteristics. The rolling deformation and heat-treatment technologies also influence the nucleation of various precipitates (κ-carbide, B2, and MC), grain size and austenite characteristics. However, the existing researches only focus on a certain scientific problem, such as the role of recrystallization annealing on MC precipitation behavior. The research and development of Fe–Mn–Al–C low-density steel is still under the gestation stage, and more systematic and in-depth researches are needed in future.

3.2. *Strain hardening mechanism*

TRIP and TWIP effects are the main deformation mechanisms for Fe–Mn–Al–C duplex phase low-density steels. When the fractions of δ-ferrite and austenite are high, the deformation mechanisms show multi-stage strain hardening characteristics. In addition, due to the absence of austenite in single ferrite Fe–Mn–Al–C steels, martensite transformation is absent. For single austenite-base Fe–Mn–Al–C steels, the deformation mechanism is closely related to stacking fault energy (SFE).

3.2.1. *Single ferrite steel*

Al content strongly influences the tensile properties of single ferrite-base Fe–Mn–Al–C steels by influencing crystal structures (Fig. 7) [80]. When the Al content reaches 6 wt%, the κ phase with a short-range ordering crystal structure is gradually formed, resulting in a sharp decrease of ductility [81]. With the increase of Al content over 7 wt%, the short-range ordering reaction continuously occurs, and the shear action of dislocations on ordered phases induces high work hardening during plastic deformation [82]. Further increasing Al content to a peak of 16 wt%, a terrific amount of B2 and DO3 phases with super crystal structures are formed, resulting in a rapid decrease of work hardening [20]. Herrmann et al. [83] observed dislocation slip bands in Fe-xAl steels, and they believed that the generation and slip of dislocations in a Fe–6Al steel determined the deformation mechanism during plastic deformation. As the Al content increased to 11–19 wt %, the number of active mobile dislocations decreased under a low deformation temperature, due to a limited number of dislocation slip systems. At this time, the number of mobile dislocations sensitively depended on crystal orientation. After fracture, the dense dislocation bundles and undeformed B2 phase were also observed in ferrite, indicating that uneven plastic deformation occurred in ferrite. The above discussion implies that the shear of mobile dislocations on ordered phases and dislocation slips mainly control the strain hardening of single ferrite steels.

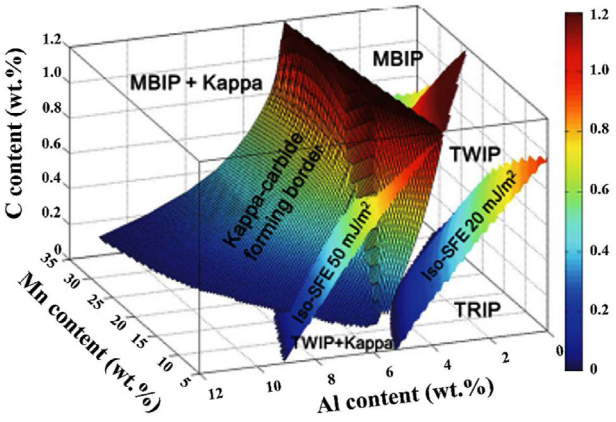


Fig. 7 – Relationship between composition, deformation mechanism and SFE in Fe–Mn–Al–C steel under room temperature [80].

3.2.2. *Single austenite steel*

SFE can significantly affect dislocation mobility during plastic deformation, thus changing the deformation mechanism. Table 3 summarizes the deformation mechanism and mechanical properties of different austenite base Fe–Mn–Al–C steels with various SFEs. Generally, it is believed that the primary deformation mechanism of austenite gradually changes from TRIP to TWIP effect and dislocation slip-induced plasticity with the increase of SFE [84]. Still, TRIP and TWIP effects are also related to dislocation slips. So, the high strain hardening ability is attributed to the volcanic dislocation slip during strain relaxation. Dislocation slip is the most basic plastic deformation mode at room temperature, which is closely related to the deformation mechanisms of TRIP, TWIP, SIP, DSBR and MBIP.

Fig. 7 shows the relationship between SFE and deformation mechanism [80]. It can be employed to predict the deformation mechanism of single austenite-base Fe–Mn–Al–C steels. The SFE values of these steels increase with the increased Al content. Thus, the deformation mechanism changes from TRIP to TWIP and then to MBIP and SIP, accompanied by κ-carbide precipitation. However, the strain hardening behavior of single austenite-base Fe–Mn–Al–C steels can not be perfectly explained only by SFE. Many scholars insisted that the influence of short-range ordered phase clusters formed via spinodal decomposition on dislocation substructure evolution was the main factor [43].

Frommeyer et al. [85] studied the tensile behavior of a Fe–28Mn–10Al-1.2C steel with the SFE of 110 mJ/m^2, and found that austenite, ferrite, and κ-carbides were coexisted in this studied steel. During the tensile deformation, deformation twins were almost restrained, and the shear band was formed accompanied by dislocation slips. At the same time, nano κ-carbides were uniformly arranged on the austenitic matrix with a certain spacing. In addition, the uniform shear deformation (namely, the SIP effect) in austenite is very beneficial to total elongation. The synergistic effect among these deformation mechanisms results in the excellent mechanical properties.

Low-density TWIP steels have a high Mn content of 20–30 wt%. After rolling, annealing and quenching treatments, the microstructure consists of complete austenite together with a large number of annealing twins. Hong et al. [86] studied the formation mechanism of deformation twins in a Fe–17.5Mn–1.5Al-0.85C TWIP steel (Fig. 8). They deemed that the prior austenite grain boundary provided the nucleated locations for deformation twins. When the tensile strain rate reached 0.02, various slip systems were activated near the austenite grain boundary. The interaction among mobile dislocations on different slip planes led to the formation of Shockley and Frank partial dislocations at positions slightly away from the prior austenite grain boundary, finally forming the stacking faults. Further straining to 0.05, deformation twins formed near the austenite grain boundary and gradually grew into the grains. Therefore, the strong dislocation movement and deformation twins systematically contributed to the high strain hardening of TWIP steels.

Park et al. [87] studied the microstructure evolution of a quenched Fe–28Mn–xAl–(0.8–1.0)C steel during tensile deformation at room temperature and proposed the MBIP mechanism. Unlike TRIP steel, the matrix microstructure of these steels was only austenite after quenching. Meanwhile, the nano short-range ordering phases were precipitated in partial areas. Under a low strain, dislocations aggregated on a single slip plane and formed Taylor crystal lattices. Upon a high strain, the microbands gradually formed due to the interaction between mobile dislocations and Taylor crystal lattices. The total dislocation density continuously increased with the increasing strain, leading to the high work hardening. At this time, the interaction among various microbands could also refine grains and enhance the strength and toughness (873 MPa tensile strength and 100% total elongation) through the dynamic Hall-Petch effect. Welsch et al. [88] studied the strain hardening mechanism of a Fe–30.4Mn–8Al–1.2C low-density steel with the SFE of 85 mJ/m^2. They acknowledged that DSBR was the main strain hardening mechanism, contributing to the excellent mechanical properties (900 MPa tensile strength and 68% total elongation). The fundamental feature of this mechanism is that the sliding band substructure is gradually refined under plastic deformation, which restrains the strain localization phenomenon in other sliding planes. However, the current research on microbands only stays in the abstract conceptual stage and does not give quantitative results.

3.2.3. *A-BDP steel*

When the C and Mn contents in the austenite steels are lower, the austenite stability decreases and δ-ferrite forms, forming the A-BDP steels. Its deformation mechanism is strongly affected by the δ-ferrite fraction, hardness difference between austenite and ferrite, and the ordered phases in ferrite.

Generally, austenite determines the deformation mechanism when the δ-ferrite fraction is less than 10%. Frommeyer et al. [85] studied the influence of solution and water-quenching

Table 3 – Comparison of the deformation mechanisms and room temperature mechanical properties of Fe–Mn–Al–C steels. TL represents Taylor lattice.

Composition (wt.%)	SFE (mJ·m^{-2})	Deformation mechanism	YS/UTS	Refs.
Fe–21Mn-1.6Al-0.11C-2.5Si	21	TWIP + TRIP	0.42	[30]
Fe–26Mn–8Al-1.5C	43	SIP + TWIP + MBIP	0.51	[80]
Fe-3.9Mn-5.8Al-0.35C	60	TWIP + SIP + TL	0.62	[16]
Fe-30.5Mn-2.1Al-1.2C	63	TWIP + SIP + DSBR	0.26	[17]
Fe–27Mn-11.5Al-0.95C	80	DSBR + MBIP	0.85	[67]
Fe–16Mn–2Al-0.45C	92	TWIP + DSBR	0.85	[58]
Fe–27Mn–12Al-0.9C	110	DSBR + MBIP	0.87	[24]
Fe–18Mn–7Al-0.85C	120	DSBR + SIP + MBIP	0.66	[19]

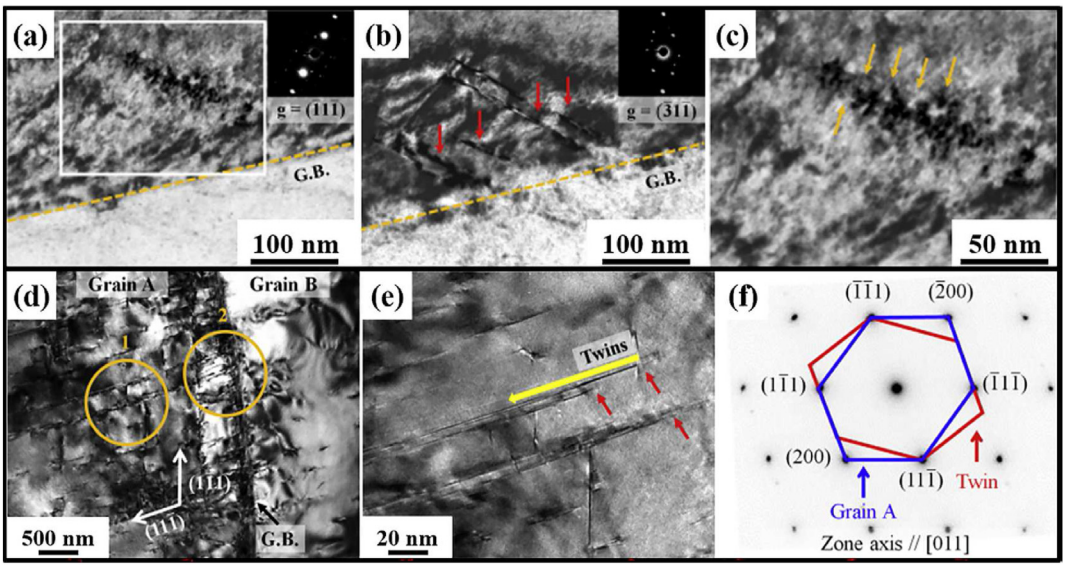


Fig. 8 – The mechanism of deformation twins near grain boundaries in Fe-17.5Mn-1.5Al-0.58C steel under various strain [86]. (a–c) After straining at 0.02, stacking faults and Frank partial dislocations on the (111) plane are indicated by red and orange arrows, respectively; (d–f) After straining at 0.05, deformation twins formed near the austenite grain boundary and gradually grew into the grains. Red arrows indicate the starting point of twins and a yellow arrow indicates the growth direction of twins in (e).

treatments on a Fe–26Mn–10Al–C steel with 883 MPa tensile strength and 65% total elongation. The uniform shear deformation of austenite determined such an outstanding combination of strength and ductility. Moreover, with the increase of δ-ferrite fraction, both ferrite and austenite provided strong strain hardening effect. Hwang et al. [89] studied the evolution of dislocation substructures in austenite and ferrite of a Fe–20Mn–9Al-0.6C duplex phase steel under different tensile strains. For austenite, the unidirectional slips were observed on the (111) plane under 5% tensile strain (Fig. 9a), and multi-directional slips occurred with the increase of strain, as shown in Fig. 9b and (c). After straining at 40%, unidirectional slips determined the deformation mechanism, while the space of slip line was much smaller than that under low strain (Fig. 9d). For ferrite, dislocations dominated the deformation mechanism at 5% low strain, and explosive dislocation multiplication developed into dislocation cells at 40% high strain (Fig. 9e and f). It can be seen that austenite shows an obvious plane slip mechanism due to the high SFE, while ferrite shows wave slips and forms the dislocation cells.

The hardness difference between ferrite and austenite in A-BDP steels results in the multi-stage strain hardening behavior during the tensile deformation [90,91]. With the increase of plastic deformation, austenite showed higher strain hardening than ferrite, so the Fe–12Mn–8Al–0.8C A-BDP steel showed three stages of strain hardening curve (Fig. 10) [90]. In

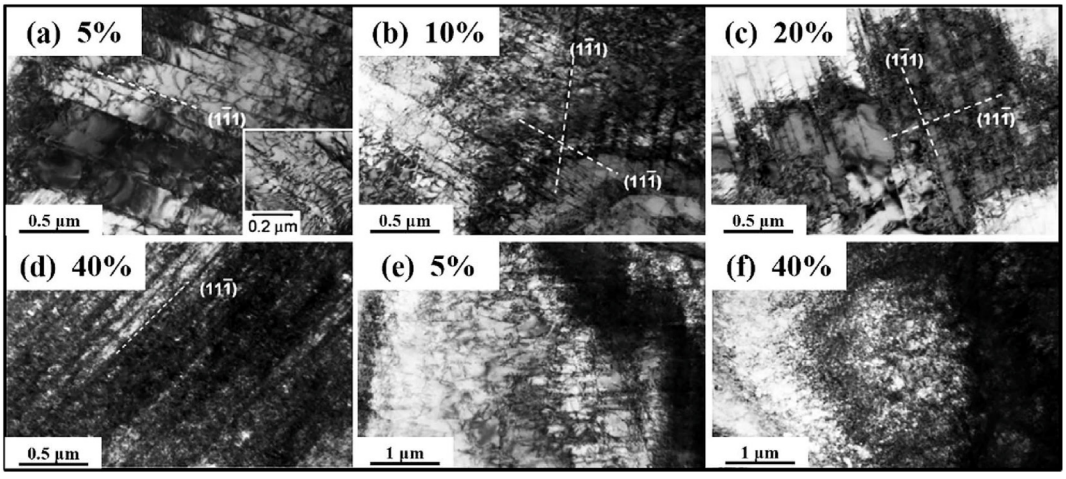


Fig. 9 – The evolution of dislocation substructures in austenite (a–d) and ferrite (e, f) of Fe–20Mn–9Al-0.6C duplex phase steel under different tensile strains [89].

the first stage, the single ferrite deformation led to a sharp decrease in strain hardening rate. In the second stage, austenite started to deform and the strain hardening curve presented approximately a line. In the third stage, the severe strain localization resulted in a quick drop in strain hardening rate, due to the obvious hardness difference between ferrite and austenite.

The ordered B2 and DO3 phases are formed on the disordered ferrite matrix of A-BDP steels, which will affect the deformation behavior [91,92]. Park et al. [91] investigated the influences of quenching and ordering annealing on microstructure and deformation mechanism of a Fe–27Mn–12Al–0.8C A-BDP steel with 27% δ-ferrite fraction (Fig. 11). The nano κ-carbide also precipitated in austenite under quenching or quenching-ordering annealing conditions, and its size gradually became larger with increasing ordering annealing temperature. Interestingly, the coarse B2 and ultrafine DO3 particles precipitated in ferrite. During tensile deformation, the κ-carbide in austenite was sheared by plane slip dislocations; however, the strong interaction among super-dislocations with narrow mechanical reverse domain boundaries occurred in ferrite. Therefore, the strain hardening of A-BDP steels is mainly controlled by the shearing effect of ordered phases by super-dislocations in ferrite and plane slip dislocations in austenite.

3.2.4. *F-BDP steel*

In F-BDP steels, the uniformly distributed ultrafine austenite on the ferrite matrix enables martensite transformation during plastic deformation [93]. However, the volume expansion caused by martensite transformation will drive the ferrite to undergo plastic deformation, resulting in the stress relaxation and transfer. The non-uniformity of plastic deformation will be significantly improved during further deformation. Therefore, the synergistic effect between ferrite plastic deformation and TRIP effect dominates the strain hardening of F-BDP steels.

Similar to A-BDP steels, the strain hardening rate curve of F-BDP steels presents various stages. Liu et al. [15] studied the strain hardening mechanism of a Fe–3.9Mn–5.8Al–0.35C

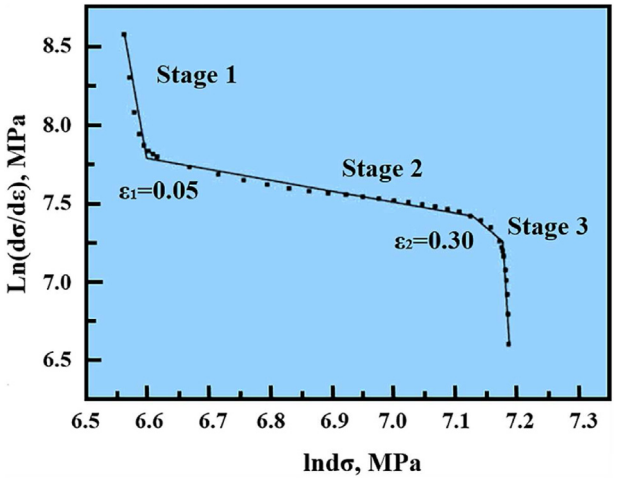


Fig. 10 – The ln(dσ/dε) vs. lnσ curves of Fe–12Mn–8Al–0.8C steel shows three stages with different slopes at the critical strains (ε_1 and ε_2) [90].

F-BDP steel under different tensile strains, as shown in Fig. 12. In the first stage, the dislocation motion in ferrite led to a sharp decrease in strain hardening rate. In the second stage, due to the low stability and fraction of austenite, the limited TRIP and TWIP effects at a low strain could not contribute much to strain hardening (Fig. 12b–d). The increase of strain hardening rate was mainly attributed to the dislocation slip in ferrite and austenite. Meanwhile, the strain hardening curve reached a peak value and entered a long-term continuous hardening stage. In the third stage, the dislocation overcame obstacles caused by the increased strain in previous stages and the dislocation motion resistance decreased, thus reducing the hardening rate (Fig. 12e and f). In the fourth stage, the martensite transformation rate was rapidly increased, and TWIP effect was also activated with the increase of strain. Therefore, the strain hardening mechanism in this stage included TRIP, TWIP effect and dislocation strengthening. In the fifth stage, the martensite transformation rate significantly decreased with the loss of austenite. Meanwhile, the volume expansion caused by martensite transformation intensified the local non-uniform deformation, resulting in a slow decrease in strain hardening rate.

The strain hardening mechanism of F-BDP steels mainly involves the TRIP and TWIP effects. Among them, TRIP effect is more crucial and closely related to the grain orientation and mechanical stability of austenite. Seo et al. [94] studied the deformation behavior of a Fe–3.5Mn–5.9Al–0.4C steel and found that the grain orientation had an important influence on austenite stability. Austenite grains with the higher schmid factor were more likely to undergo martensite transformation. In addition, the previous investigations showed that the austenite stability was mainly affected by: (1) grain size, namely, the coarser the austenite grains, the poorer the austenite stability [95]; (2) C content in austenite, namely, the larger C content in austenite, the higher the austenite stability [96]; (3) constraining effect from the adjacent phases surrounding austenite [97]; and (4) austenite volume fraction [98]. However, there are still disputes about these factors. For example, the coarse austenite grains near martensite were more stable than ultrafine austenite with lath-like or film morphologies [99]. These factors are interrelated and jointly determine the austenitic stability. The evaluation criteria of these influencing factors should be further clarified in future. For example, C content in austenite is the influential factor determining its stability, which has been reported in medium Mn steels [100].

From the previous investigations, it can be found that the strengthening mechanisms of solution strengthening, grain refinement strengthening, precipitation strengthening, dislocation slips, and strain hardening are displayed in the above-mentioned four kinds of Fe–Mn–Al–C low-density steels. Therein, the strain hardening mechanisms of both plane slip dislocations and shear effect of mobile dislocations on the ordering phases are presented in all the above steels. Still, the main strain hardening mechanism among them is slightly different. In the single austenite steels, MBIP, DSBR, SIP, and TL play a great role, while that in A- and F-BDP steels is mainly determined by TRIP and TWIP effects. Moreover, the deformation mechanisms in A- and F-BDP steels also exhibit multi-

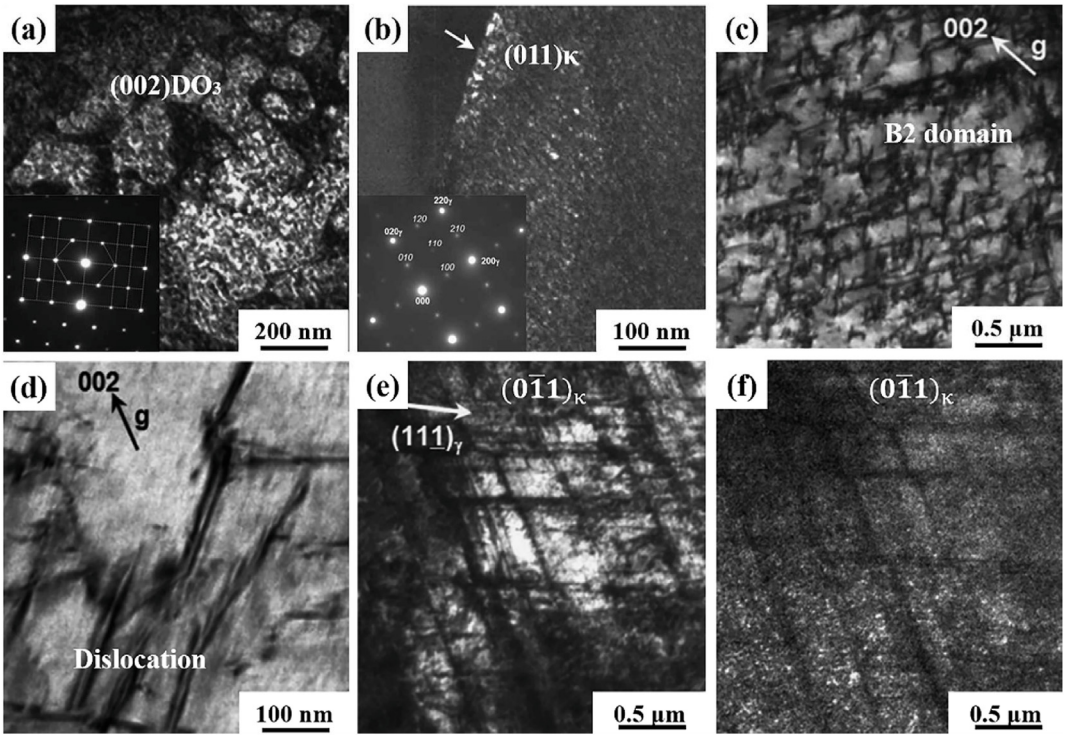


Fig. 11 – Microstructure of ferrite and austenite of the as quenched-annealed Fe–27Mn–12Al–0.8C duplex steel before and after tensile deformation [91]. (a) Fine DO3 particles in ferrite; (b) Nano κ-carbide in austenite; (c, d) Deformed microstructure of ferrite 5% strain, dislocations inside the B2 domain; (e, f) Deformed microstructure of austenite 5% strain.

stage strain hardening behavior when the fractions of δ-ferrite and austenite are high. The coupling and competition of various strain hardening mechanisms make the deformation extremely complicated, especially in the dual-phase Fe–Mn–Al–C steels. Therefore, it is necessary to further elucidate the coupling and competition mechanisms via more advanced characterization equipment in future.

4. Application performance of Fe–Mn–Al–C low-density steels

4.1. *Cryogenic performance*

For Fe–Mn–Al–C low-density steels, the austenite matrix can provide high work hardening to resist the damage under

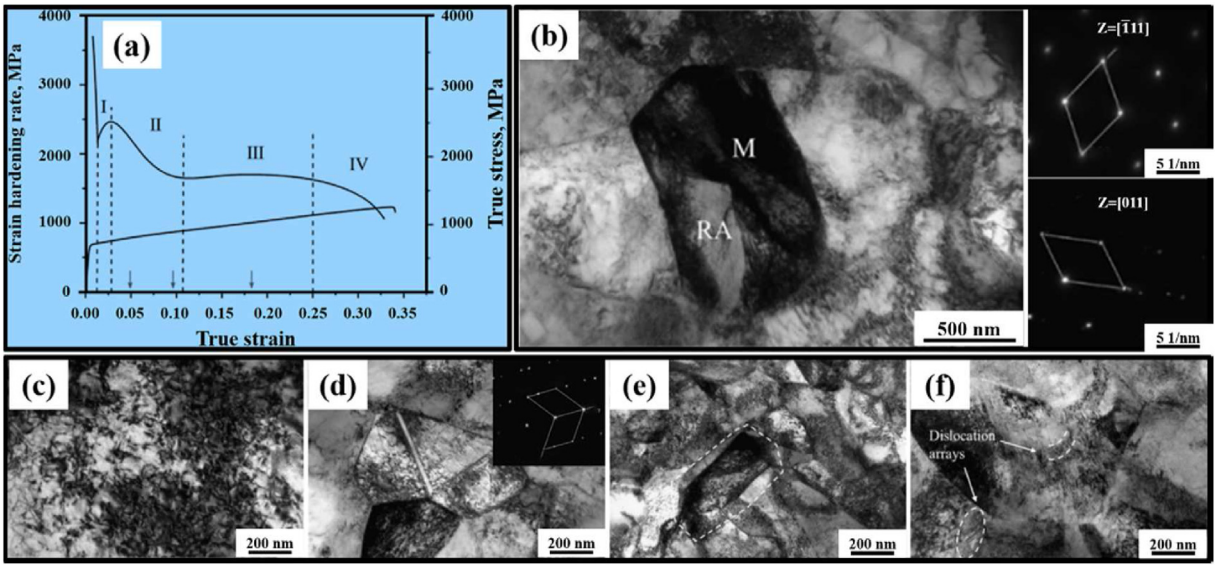


Fig. 12 – Strain hardening rate curve (a) and the corresponding strengthening mechanism (b–f) of Fe–3.9Mn–5.8Al–0.35C F-BDP steel under different tensile strains [15]. (b–d) Straining at 10%; (e, f) Straining at 20%.

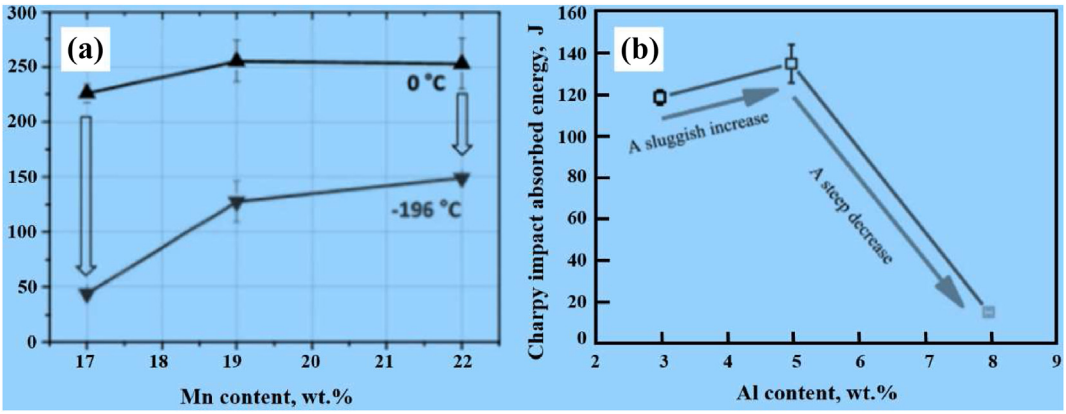


Fig. 13 – (a) Charpy impact absorbed energy of Fe-xMn-2Al-0.45C-0.05Si steel as a function of Mn content [101]; (b) Charpy impact absorbed energy of Fe–18Mn–xAl-0.6C steel as a function of Al content [102].

extreme conditions. With the rapid development of natural gas and hydrogen energies, the cryogenic performance of metal materials needs to be improved so as to satisfy the storage and transportation of above gases. Note that the Fe–Mn–Al–C low-density steels are considered one of the most promising structural material, its cryogenic performance is necessary to investigate.

The element content strongly influences cryogenic toughness. Lee et al. [101] pointed out that the (17, 19 and 22 wt%) Mn addition into Fe-xMn-2Al-0.45C low-density steels significantly affected the tensile and Charpy impact properties under cryogenic conditions (Fig. 13a). However, the influences of Mn content on the tensile and Charpy impact properties at room temperature were minimal. Under cryogenic conditions, the ductility and impact energy gradually decreased with the increase of Mn content. The corresponding cryogenic ductility for 17, 19 and 22 wt% Mn steels were 75.7%, 53.4% and 9.7%, respectively. Whereas the corresponding impact energies were 80.5, 49.8 and 41.1 J, respectively. Moreover, Chen et al. [102] investigated the effect of (3, 5 and 8 wt%) Al addition into Fe–18Mn–xAl-0.6C low-density steels on impact energy at cryogenic temperature (Fig. 13b). With the Al content increasing from 3 to 5 wt%, the impact energy at cryogenic temperature increased from 122 to 138 J. However, 8 wt% Al content resulted in a decrease of impact energy at cryogenic temperature.

The secondary phase transformation is significant for the deterioration of the impact energy at cryogenic temperature. For example, the strain-induced martensite transformation during plastic deformation could relieve the local stress and enhance the impact energy of a Fe–18Mn–8Al–0.6C low-density steel [102]. The coarse banded δ-ferrite phase boundaries acted as the center of microcrack nucleation and propagation during tensile or impact deformations, finally resulting in the unacceptable degradation of mechanical properties at cryogenic temperature. In addition, the microband structure can also enhance the mechanical properties of Fe–Mn–Al–C low-density steels at cryogenic temperature. Gutierrez-Urrutia et al. [103] systemically investigated the microband evolution in a Fe–30Mn–6.5Al–0.3C (wt.%) low-density steel during tensile deformation at −196 °C (Fig. 14). Under a low strain, the intense movement of closely spaced slip bands activated the planar slip localization, finally forming the microbands (Fig. 14a and b). At a high strain, the new

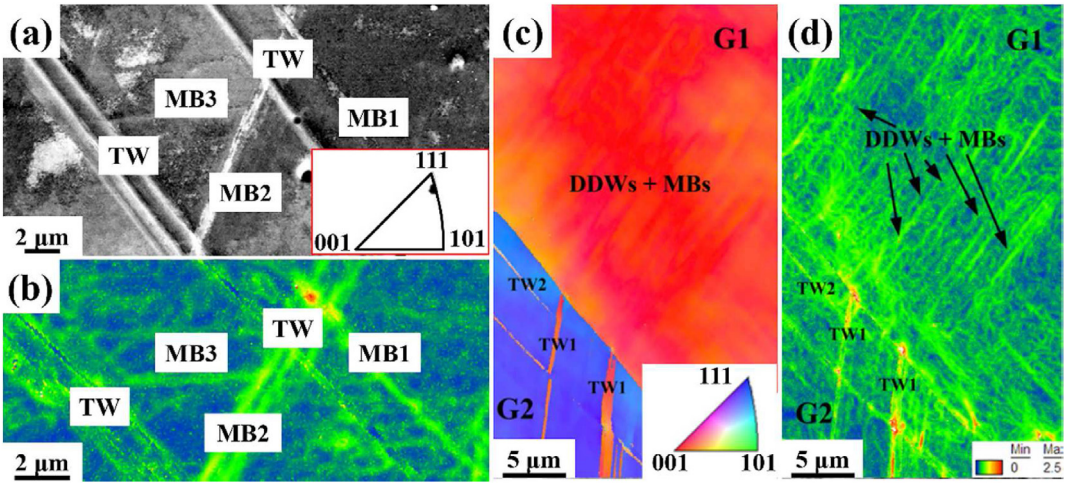


Fig. 14 – The microstructure evolution in Fe–30Mn–6.5Al–0.3C (wt.%) low-density steel during tensile deformation at −196 °C [103].

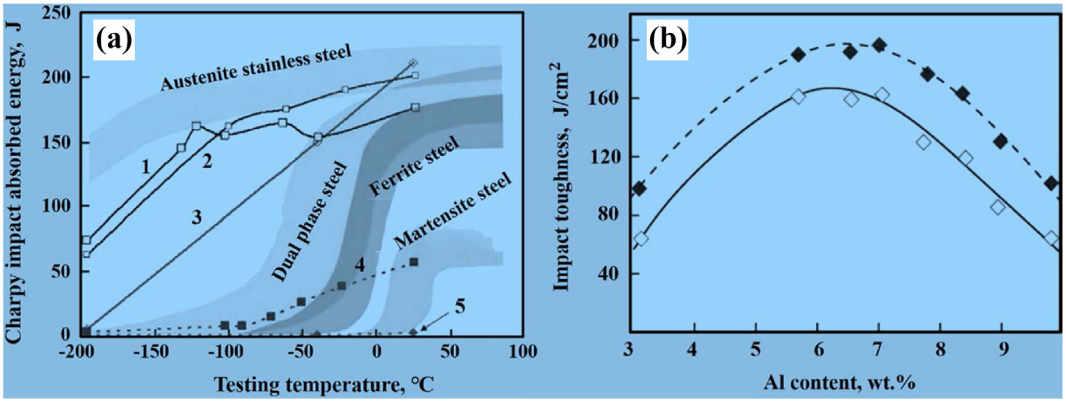


Fig. 15 – (a) Relationship between Charpy impact absorbed energy and testing temperature for Fe–Mn–Al–C austenite steel [104,105]. (b) Relationship between impact toughness and Al content for Fe–Mn–Al–C steel [106].

microbands formed together with the generation of deformation twins (Fig. 14a and b). However, the microbands located within austenite grains were not conducive to the formation of deformation twins (Fig. 14c and d). This complex deformation structure led to the highly uneven deformation behavior, which significantly enhanced the work hardening and mechanical properties of studied steels under cryogenic conditions.

4.2. *Impact toughness*

Impact toughness is one of the essential properties for evaluating automotive steels, representing the ability to absorb plastic deformation and fracture energies under impact loading. The impact toughness of Fe–Mn–Al–C steels depends on the phase composition and its distribution, especially the κ-carbide precipitation and δ-ferrite formation [104].

Fig. 15a shows the relationship between the Charpy impact absorbed energy of Fe–Mn–Al–C A-BDP steels and testing temperature [104,105]. It can be seen that the impact energy of A-BDP steels after solution treatment was much higher at room temperature, even exceeding the typical austenite stainless steel at room temperature (as shown in curves 1, 2 and 3 in Fig. 15a). When the testing temperature was below −100 °C, there was an obvious ductile-brittle transition stage. After aging at 550 °C for 16 h, the impact energy of Fe–Mn–Al–C A-BDP steels at room temperature decreased significantly (as shown in curves 4 and 5 in Fig. 15a), especially the value of a Fe–30Mn–10Al–1C steel is close to 0 J. This was mainly related to the precipitation of κ-carbide.

Al content also affects the impact toughness of aged Fe–Mn–Al–C steels (Fig. 15b) [106]. This phenomenon is closely related to κ-carbide precipitation and δ-ferrite formation. When Al content was lower than 7%, κ-carbide could not precipitate, but Fe_3C_7 and M_7C_3 might precipitate along grain boundaries during aging, reducing the impact toughness. After increasing Al content to 7%–10%, the formations of Fe_3C_7 and M_7C_3 were inhibited, but the brittle κ-carbide gradually precipitated along the grain boundary, also leading to a decreased impact toughness. When Al content exceeded 10%, δ-ferrite gradually formed and further reduced the impact toughness. Due to the deformation incompatibility between austenite and ferrite phases, microcracks were mainly generated at the phase interface.

In short, the Al element affects the precipitation position and particle size of κ-carbide, thereby affecting the impact toughness of Fe–Mn–Al–C steels. In contrast, the impact toughness of Fe–Mn–Al–C austenite steels is higher under the conditions of solution treatment and water quenching than that under aging treatment. In addition, due to the formations of short-range ordering phase and κ-carbide, the impact toughness is significantly affected by the cooling rate after solution treatment.

4.3. *Fatigue performance*

Fatigue fracture is one of the main failure modes in metallic components under variable loading, thus numerous studies about fatigue behavior of Fe–Mn–Al–C steels have been carried out. Kalashnikov et al. [107] reported that the fatigue life of a Fe–28Mn–9Al-0.86C austenite low-density steel was about twice that of a Fe–12Cr–1.5Ni–0.2V–1.8W–0.5Mo–0.15C martensite stainless steel. Hamada et al. [108] studied the fatigue crack growth behavior of Fe–Mn–Al–C austenitic low-density steels. They found that this steel had a stronger ability to prevent fatigue crack growth than that of 304 L austenite stainless steels. However, due to the limitations of objective factors and test data, the fatigue behavior of Fe–Mn–Al–C austenite low-density steels is still unclear.

Hamada et al. [109] systematically investigated the fatigue behavior of high-Mn steels with various grain sizes. They insisted that the decreased grain size enhanced the fatigue resistance, and the cracks preferentially generated at sites of intersections of slip bands, grain boundaries, and annealing twin boundaries. However, the continuous slip bands were generated and deformation twins were absent during high-cycle straining [109]. Subsequently, they analyzed the cyclic behaviors of a Fe–16Mn–1.5Al–0.3C steel with fine- and coarse-grained structures [110]. In the coarse-grained steel, deformation twins, stacking faults, planar dislocation arrays, and dislocation tangles were observed under cyclic loading. Still, the deformation twins were only displayed in the fine-grained steel (Fig. 16a and b). It can be seen that the fatigue behavior of Fe–Mn–Al–C steels are closely related to

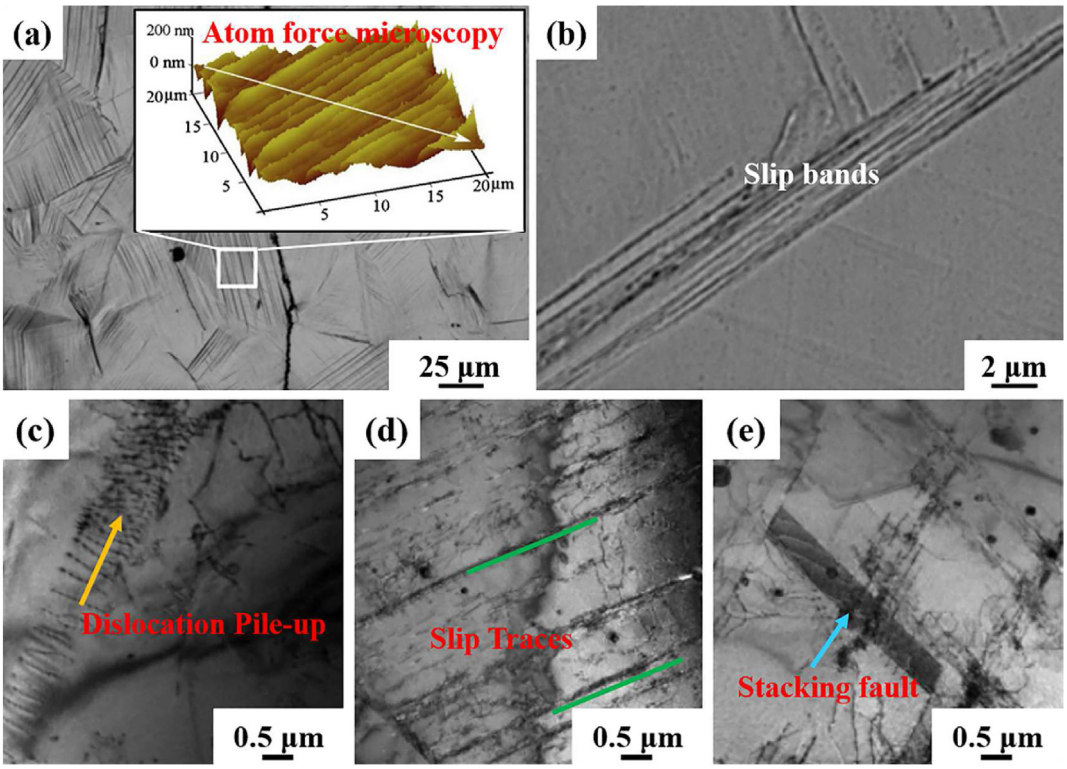


Fig. 16 – The fatigued surface and atom force microscopy images (a) and slip band structure (b) in Fe–16Mn–2Al–0.3C steel [109]. The deformed microstructures of micro-alloyed Fe–28Mn–7Al–1C steel after fatigue failure at 0.4% (c, d) and 0.8% (e) strains [111].

chemical compositions and microstructures. Zhao et al. [111] pointed out that the microalloyed element additions in Fe–Mn–Al–C steels exhibited a higher fatigue life than that of non-microalloyed one. Numerous parallel deformation bands together with planar dislocations and nano-size precipitates showed a positive influence on enhancing the fatigue life (Fig. 16c–e). The previous reports mainly focus on the effect of various deformation mechanisms on fatigue behavior of Fe–Mn–Al–C steels. However, the influence of micro-alloyed carbides on the cyclic behavior of Fe–Mn–Al–C steels is reported minimally.

4.4. *Hydrogen embrittlement*

Hydrogen embrittlement is the hydrogen dissolved in steels, which causes stress concentration and then forms microcracks, resulting in catastrophic fracture of the structural material and component [112–115]. Once hydrogen embrittlement is generated, it can not be eliminated. Fe–Mn–Al–C low-density steels are considered as high-strength steels for industrial applications. It is necessary to study the hydrogen embrittlement sensitivity and its influence on mechanical properties. Koyama et al. [112] insisted that the fracture modes induced by hydrogen embrittlement were mainly composed of intergranular and transgranular fractures in a Fe–18Mn–1.2C steel. Meanwhile, the transgranular fracture was initiated parallel to the primary and secondary deformation twin boundaries, resulting in the generation of cracks at grain boundaries and twin boundaries. Afterwards, they examined the hydrogen-assisted cracking mechanisms of a Fe–26Mn–11Al–1.2C steel [113]. They proposed that the grain boundary triple junction cracking and slip-localization-induced intergranular cracking along micro-voids determined the embrittlement ability. Meanwhile, grain boundary triple junction cracking was preferentially generated on grain boundaries, while the microscopically ductile slip-localization-induced intergranular cracking assisted crack growth during plastic deformation, resulting in the macroscopic brittle fracture appearance (Fig. 17).

In addition, the precipitates (κ-carbide, B2, DO3, and β-Mn et al.) also closely act with hydrogen and affect the hydrogen embrittlement of Fe–Mn–Al–C steels [114]. Due to the high SFEs, numerous κ-carbides are often precipitated. Thus, the hydrogen embrittlement is closely related to κ-carbide particles [115]. It is reported that C content in Fe–Mn–Al–C steels determined the hydrogen embrittlement [114]. For high C (ω_C > 1 wt%) steels, hydrogen embrittlement occurred in the interdendritic area due to the presence of κ-carbide, while the κ-carbide could limit crack propagation. For low C (ω_C < 0.05 wt %) steels, the hydrogen embrittlement occurred in DO3 phase. Given the chemical formula composition and crystal structure characteristics of κ-carbide, hydrogen is easy to combine with it to form a relatively stable hydride (Fe_3AlH). Based on the density-functional theory, Timmerscheidt et al. [115] calculated the interaction between κ-carbide and hydrogen dissolving. The interaction of H–H showed a strong repulsive force in a short distance, while C–H showed a strong repulsive force in a long distance. In addition, the efficiency of κ-carbide

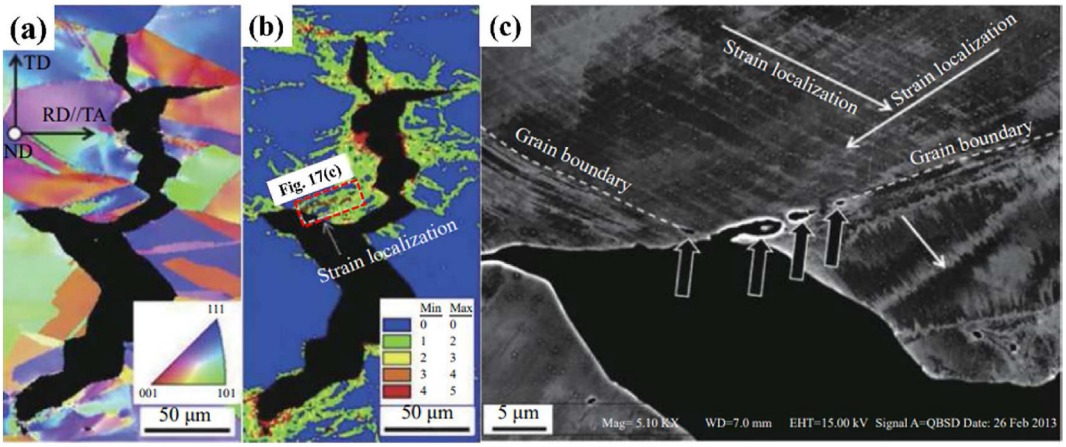


Fig. 17 – Hydrogen-induced intergranular crack in Fe–26Mn–11Al–1.2C austenitic steel containing κ-carbides [113]. (a) Inverse pole figure. (b) KAM figure. (c) Orientation-optimized ECC image corresponding to the part surrounded by the red broken lines in (b).

capturing and diffusing hydrogen depended on C and Mn contents. C content determined the number of κ-carbide and thus the capacity of capturing hydrogen. However, Mn content determined the overall efficiency of capturing hydrogen. Therefore, to control hydrogen diffusion in κ-carbide, the C content in κ-carbide should be increased while the Mn content should be reduced. Moreover, the presence of vacancies at the κ-carbide/austenite interface could improve hydrogen solubility. So, the κ-carbide/austenite interface would consider as an effective hydrogen capture center.

4.5. *Wear performance*

Due to the interaction among various strengthening mechanisms, Fe–Mn–C–Al low-density steels show good wear performance. Sohrabizadeh et al. [116] evaluated a relationship between aging time and wear rate in a Fe-22.33Mn-1.35C-9.87Al steel. After aging at high temperature or for long duration, the κ-carbide within austenite matrix could inhibit crack nucleation during the wear test. Peng et al. [117] selected a quenched Fe–25Mn–1.3C–6.6Al steel upon aging at 550 °C for different time (from 1 to 4 h) to conduct impact abrasive wear tests (Fig. 18). The fine nano-size κ-carbide precipitated coherently within the austenite matrix under the aging treatment for 2 h (Fig. 18a), which resulted in the best wear resistance and hardness (Fig. 18e). However, a longer aging time resulted in the decomposing of austenite into κ-carbide toward the whole austenite grains and grain boundaries (Fig. 18b). These over-coarsened κ-carbide along grain boundaries made the material extremely brittle, decreasing the wear resistance (Fig. 18e). The surface hardening showed Lomer–Cottrell locks, high-density dislocation tangles, and dislocation walls; neither martensite nor deformation twins were observed under the low impact energy condition (Fig. 18c and d). It can be seen that the aging time play an essential role in the wear resistance of Fe–Mn–C–Al low-density steels. The suitable aging time and temperature should be further studied.

In recent years, the metal injection molding process has been proposed further to optimize the wear-resistant

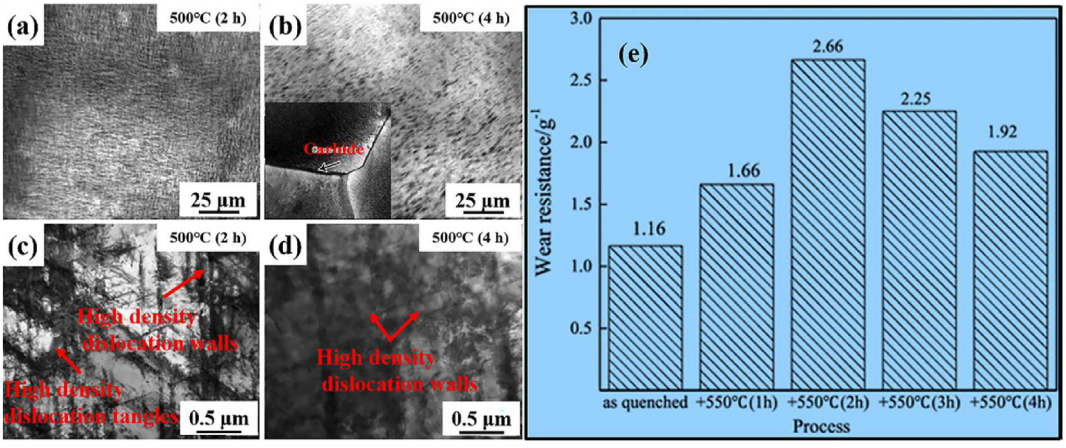


Fig. 18 – Microstructure (a–d) and wear resistance (e) of Fe–25.1Mn–6.6Al–1.3C steel under quehching at 1050 °C and aging at 550 °C for 2 and 4 h [117].

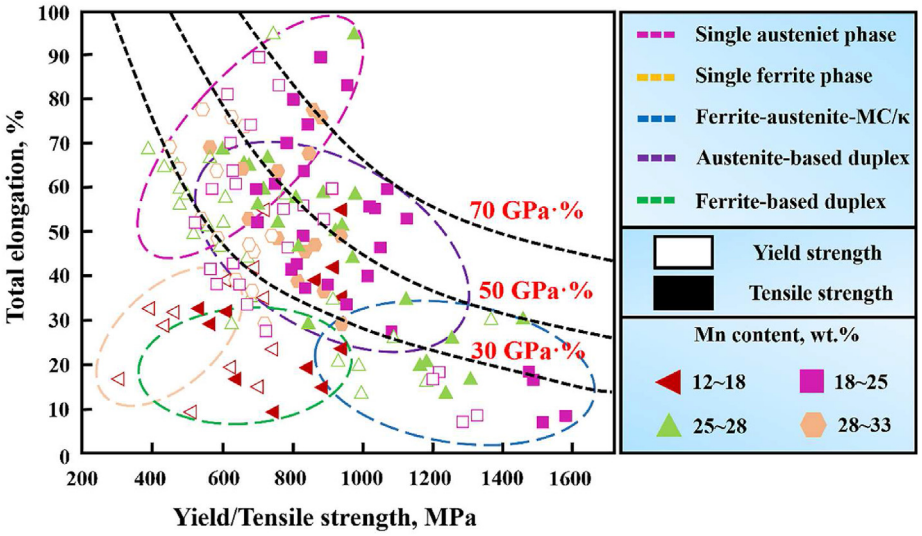


Fig. 19 – The measured room-temperature tensile properties of Fe–Mn–Al–C low-density steels with various Mn contents [15,23,34,51,52,79,80], [91–93], [120–150].

properties of Fe–Mn–C–Al low-density steels. Garcia-Aguirre et al. [118] mixed a Fe–22Mn–0.4C–1.5Al–1.5Si steel powder with high-density polyethylene and paraffin wax and immersed them in the solvent. It was then annealed between 1200 and 1360 °C to obtain full austenite with uniformly distributed MnO_2 particles. According to this manufacturing method, superhard SiC, Ni_xAl_y (NiAl, Ni_3Al and Ni_2Al_3) and $(Fe, Mn, Cr)_3C$ particles can be mixed with steel powders to prepare composite materials, further improving the wear-resistant properties of Fe–Mn–C–Al steels. For example, El-Bagoury et al. [119] studied the influence of alloying elements and solution treatment conditions on the microstructure and its turn-on wear resistance of Fe–Mn–Al–C–Cr steels. After solution at 1050 °C for 2 h, the blocky or lath-like $(Fe, Mn, Cr)_3C$ precipitates were observed besides austenite, ferrite, and martensite. With the increase of solution time to 4 h, the partial $(Fe, Mn, Cr)_3C$ precipitates were dissolved, and the matrix microstructure became full austenite. Therefore, the low volume fraction of $(Fe, Mn, Cr)_3C$ precipitates and high volume fraction of austenite lowered the wear resistance of the studied steel. In addition, as the C, Mn and Cr contents increased, the volume fraction of $(Fe, Mn, Cr)_3C$ precipitates increased, which could enhance the wear resistance. Note that the metal injection molding process is a promising method to enhance the wear resistance of Fe–Mn–Al–C low-density steels. However, the addition of micro-alloying elements with excessively high mass fraction conflicts with the development of low-cost high strength steels. Therefore, it is necessary to develop the injection molding process of Fe–Mn–Al–C low-density steel without micro-alloying elements in future.

4.6. *Tensile properties*

Fig. 19 shows room-temperature the tensile properties of Fe–Mn–Al–C low-density steels with various Mn contents. [91–93], [120–150]. It is clearly indicated that the relationship between strength and ductility of Fe–Mn–Al–C steels follows the typical "banana" curve, the mechanical properties of single austenite and A-BDP steels seems to be superior than those of other ones. For example, the Fe–30Mn–11Al–1.2C steels with single austenite phase shows 1058 MPa yield strength, 1310 MPa tensile strength, and 38% total elongation [137]. Both the fine intragranular κ-carbide and intergranular one can significantly enhance the yield strength without significantly deteriorating the ductility but result in the increased yield ratio (>0.8) of Fe–Mn–Al–C low-density steels. Moreover, a high Mn content seems to expand more space for enhancing the strength and ductility of Fe–Mn–Al–C steels, but it also can bring the increased material cost and processing difficulty. Achieving good performance and ensuring good economy are the long-term targets during its sustainable development of Fe–Mn–Al–C low-density steels.

5. Conclusion and prospect

Under the target demand of "C neutrality" and "C neutralization", the Fe–Mn–Al–C low-density steel will play an essential role in fuel savings. Due to its outstanding mechanical properties, impact resistance and hydrogen embrittlement resistance, and wear performance, it is widely used in automobile, military and construction fields. However, there are still numerous scientific problems to be further studied. In addition, basic research on the service properties of Fe–Mn–Al–C low-density steels is insufficient. The investigation of Fe–Mn–Al–C low-density steels can be started from the following aspects.

(1) Alloy design. The variations of Mn, Al, and C contents lead to the difference of phase constituents in Fe–Mn–Al–C low-density steels and a wide range of tensile properties can be acquired. Still, Al addition can achieve the weight saving and microstructure

regulation, the limitation of Al addition should be further adequately clarified to exploit its lightweight role. The Mn, Al and C additions should be also tailored properly to exert their advantage role in enhancing the performance. In addition, the additions of micro-alloying elements (Nb, V, Ti, Cr, and Ni et al.) need to be explored so as to optimize the performance and thereby broaden the service fields of Fe–Mn–Al–C low-density steels.

(2) Smelting process. High Mn and Al contents result in the poor heat dissipation rate of steel ingot and Al-oxide pollution during the casting process, respectively. A small difference in element content can lead to significant variations of precipitate types, especially in A- and F-BDP steels. The depth-type purification of metallurgical is the primary challenge to solving the Mn-and Al-enriched problems.

(3) Thermal mechanical processing. Adopting the combination between traditional hot/cold rolling and heat-treatment to achieve comprehensive properties of Fe–Mn–Al–C steels is feasible. However, the sensitivity of properties to temperature and time easily results in a series of problems in the industrial production lines. It is necessary to explore and develop new processing technologies. For instance, powder metallurgy could be a promising method for manufacturing the Fe–Mn–Al–C low-density steels.

(4) Strain hardening mechanism. The competition and coupling among various strain hardening mechanisms increase the difficulty for analyzing the deformation behavior of Fe–Mn–Al–C steels. Moreover, the stress/strain partitioning among various interfaces also strongly affect microstructure evolution and deformation behavior. Simulation methods should be helpful for analyzing the above-mentioned laws and problems.

(5) Service properties. Research work on the influence of precipitation behavior on service properties of Fe–Mn–Al–C low-density steels is still limited. Especially, the influence of short-range order and κ-carbide on service properties is still a debatable point. Systematic research on the service properties of Fe–Mn–Al–C lightweight steels is also needed.

In conclusion, the comprehensive properties of Fe–Mn–Al–C low-density steels can be optimized by tailoring the composition design, controlling the type, size and distribution of various precipitates in the matrix, and optimizing the technology parameters of thermal-mechanical processing to meet the "strength, toughness, and lightweight" index and further expand its application areas.

Credit author statement

Shao-bin Bai was responsible for writing and editing the manuscript. Yong-an Cheng, Xin-Liu, and Hui-hu Lu searched the literatures. Pei-kang Bai, Da-zhao Li, Zhi-quan Huang, and Jing-yang Li were responsible for modifying this manuscript.

Declaration of competing interest

We declare that we have no conflicts of interest.

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